Project Summary: Employee Attrition Analysis

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The Problem

Employee turnover is a costly challenge for any organization, leading to lost productivity, high recruitment expenses, and the loss of institutional knowledge. Our HR team faces the difficulty of identifying high-risk employees *before* they resign. Currently, without a predictive approach, we are often reacting to resignations rather than preventing them, which makes it hard to maintain team stability and focus on long-term retention.

Overview

In this project, I developed a machine learning model to predict employee attrition. By analyzing historical workplace data, I aimed to uncover the hidden patterns behind why employees leave, helping the HR team transition from a reactive approach to a proactive, data-driven retention strategy.

My Findings

- **Which factors influence attrition the most?**
  My analysis identifies **Overtime**, **Monthly Income**, and **Age** as the most significant drivers of employee turnover.

- **How accurate was your model (in plain terms)?**
  The Gradient Boosting model performed best, achieving an **AUC score of 0.79**, indicating a strong capability to distinguish between employees who will likely stay and those who are at risk.

- **What surprised me?**
  While compensation is important, I was surprised to find that work-life balance—specifically the requirement for excessive overtime—is equally critical, often acting as the tipping point for resignation.

Recommendation for Business

If I were advising the HR Director, I would recommend focusing on:

- **Workload Review Policy**: Capping excessive overtime, especially for high-turnover roles like Sales Representatives and Laboratory Technicians.

- **Retention Dialogue Program**: Using model insights to identify high-risk employees for supportive, one-on-one check-ins before they reach the point of resignation.

A Note on Reality

It is important to note that this model predicts **probability, not certainty**. It flags patterns based on historical data, so it should be used to guide human conversation rather than as a final decision-making tool.